AI & MACHINE LEARNING CAPSTONE PROJECT

AI-Powered Customer Churn Prediction & Behavioral Segmentation Microservice

Anend-to-end machinelearningpipelinefromraw customer data toproductionFastAPI serving.

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Python | Scikit-learn | FastAPI | Joblib | K-Means | Random Forest



THE BUSINESS CASE

Retention is the next growth lever

Thechallenge: Acquisitioncosts 5–7× morethan retention,while reactivesupport leaves recurring revenue exposed.

The opportunity: Move from reactive service to proactive, predictive retention.

Project goal: Identify at-risk accounts 30–60 days before churn and recommend targeted incentives.

30–60

Days before churn to identify at-risk accounts

≥75%

<100 ms

DATA FOUNDATION

A leak-free pipeline turns messy accounts into model-ready signals

Ingest 7,043 customer accounts and 21 raw attributes spanning demographics, service usage, and charges.	**Clean** Convert hidden whitespace values in TotalCharges to missing; impute with the median; drop customerID noise.
Engineer One-hot encode categorical variables with drop_first=True to avoid multicollinearity.	**Normalize** Fit StandardScaler inside strict train-test boundaries to prevent leakage.

SUPERVISED MODEL BENCHMARKING

Random Forest wins the first benchmark

F1 0.60 ROC-AUC 0.84

Ensemble bagging reduces variance

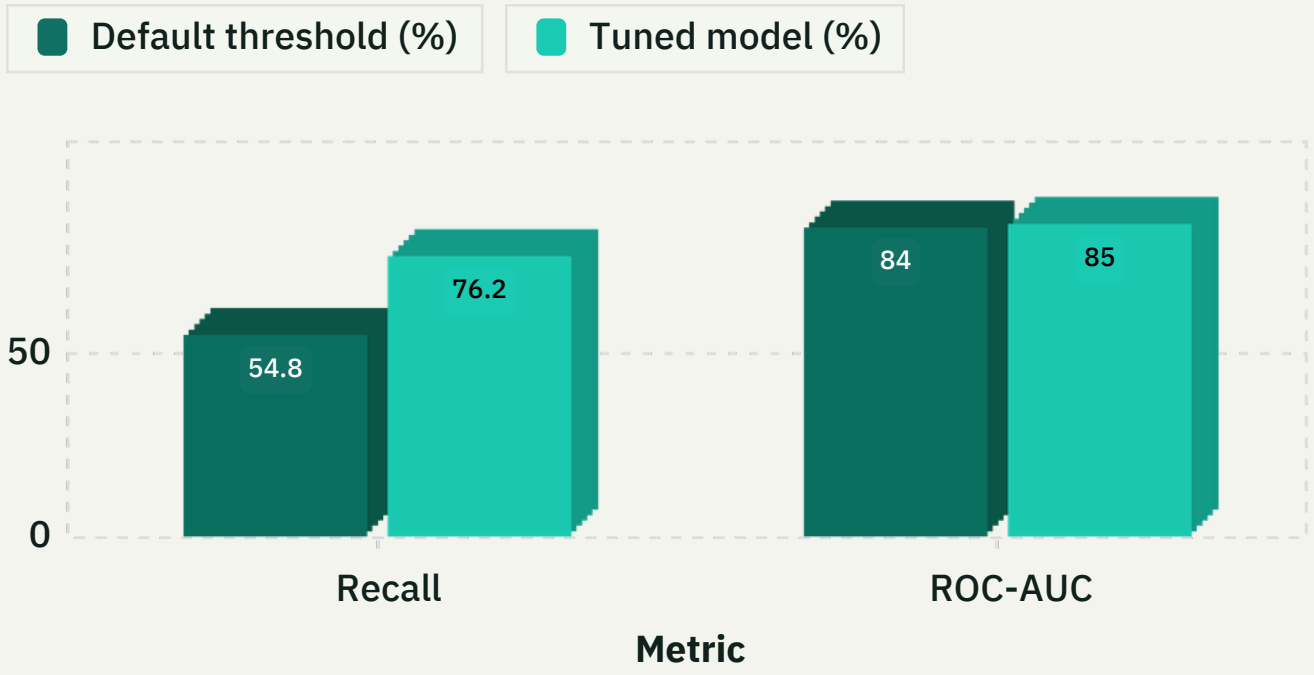
andcaptures non-linear interactions better than a single tree or distance-based baseline.

Model	Accuracy	F1	ROC-AUC
Logistic Regression	80.0%	0.59	0.84 0.82
Decision Tree	78.8%	0.56	**0.84**
Random Forest (Champion)	80.2%	**0.60**	

The champion delivers the strongest baseline. Decision Tree carries higher overfitting risk and rigid axis splits.

MODEL TUNING & RESOLVING IMBALANCE

Tuning shifts the model from average accuracy to useful intervention



THE INTERVENTION LENS

+21.4 points

Recall rises from 54.8% to 76.2%.

73.5% non-churn versus 26.5% churn caused default thresholds to miss 45% of churners.

ROC-AUC reaches 0.85, with fewer costly false negatives.

5-fold stratified cross-validation with GridSearchCV tunes n_estimators, max_depth, min_samples_split, and class_weight='balanced'.

BEHAVIORAL SEGMENTATION

Three behavioral personas make retention actions specific

PCA for 2D variance mapping, then Elbow Method and silhouette scores across K=2–8.



High-Risk New Subscribers

Tenure <12 months, month-to-month contracts, churn >45%.

Action: onboarding check-ins and first-year discounts.



High-Value Champions

Tenure >40 months, bundled fiber and streaming, churn <10%.

Action: VIP loyalty rewards and proactive account care.



Low-Touch Budget Users

Moderate tenure, basic phone/DSL plans, low spend.

Action: flexible digital billing and automated payment options.

PRODUCTION SERVICE & REAL-TIME API

FastAPI turns the pipeline into a real-time decision microservice

Low-latencyinference delivering predictive scores and automated retention actions in under 100ms.

<100 ms

Infrastructure

- Bundle Random Forest, StandardScaler, and K-Means into one Joblib binary
- GET /health
- POST /predict
- Pydantic schemas and fallback defaults

Live inference payload

Input


```
{"tenure": 2,
"MonthlyCharges": 95.0,
"Contract": "Month-to-month"}
```


Output


```
{"churn_prediction": "Churn", "probability": 0.82, "risk_level": "High", "assigned_cluster": "Cluster 0", "recommended_action": "Urgent retention outreach"}
```

THE PATH FORWARD

From four-week capstone to a scalable retention operating system

Delivered

An end-to-end lifecycle from exploratory analysis through a live, tested prediction API.
Business impact
Retention teams move from guessing to automated, data-driven customer preservation.

Next

Containerise with Docker, deploy to AWS/GCP, monitor data and concept drift, and connect Salesforce/HubSpot to trigger retention workflows.

